

Programmable Logic Devices

2.2.2 Programmable logic devices and gate arrays

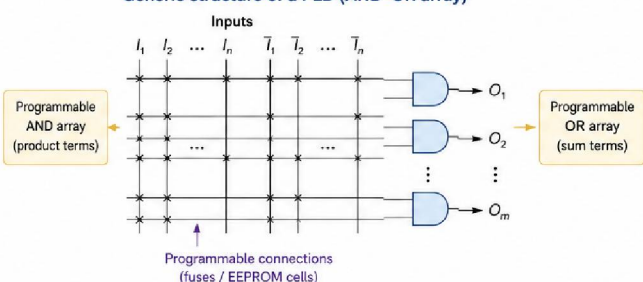
2.2.2 Programmable logic devices and gate arrays

Programmable logic devices (PLDs) and gate arrays are integrated circuits whose logic functions can be configured or customized after manufacturing.

1. Programmable Logic Devices (PLDs)

- Highly flexible devices whose logic functions can be programmed by the user.
- Programming is typically non-volatile (e.g. EEPROM, Flash).
- Examples: PROM, PAL, GAL, CPLD, FPGA.

Generic structure of a PLD (AND-OR array)



Examples of PLDs

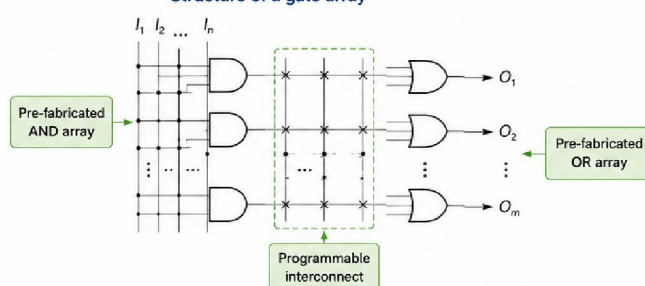
Device	Description
PROM	Programmable ROM – fixed OR array, programmable connections.
PAL	Programmable Array Logic – programmable AND, fixed OR.
GAL	Generic Array Logic – like PAL, reprogrammable (EEPROM).
CPLD	Complex PLD – multiple PLD blocks with interconnect.
FPGA	Field-Programmable Gate Array – very high density and flexibility.

Summary: PLDs provide high flexibility and are reprogrammable, ideal for development and low-volume applications. Gate arrays offer higher speed and lower cost for large-volume, dedicated applications.

2. Gate Arrays

- A gate array is a semi-custom IC where the AND and OR arrays are fabricated on the chip, but the interconnections are customized.
- Less flexible than PLDs but faster, cheaper and lower power for high-volume applications.
- Used in ASICs and structured designs.

Structure of a gate array



PLDs vs Gate Arrays

Feature	PLDs (e.g., CPLD, FPGA)	Gate Arrays
Flexibility	Very high (user programmable)	Lower (semi-custom)
Performance	Lower (more interconnect delays)	Higher (customized interconnect)
Cost	Higher per unit	Lower per unit (high volume)
Design effort	Low to medium	High (requires customization)
Typical use	Prototyping, low/medium volume	High-volume production

now explain

2.2.2 Programmable Logic Devices (PLDs) and Gate Arrays

English + Nepali Mix (No Romanize)

Introduction

Digital systems design गर्दा logic circuits implement गर्न विभिन्न प्रकारका integrated circuits (ICs) प्रयोग गरिन्छ।

Traditional ICs जस्तै SSI, MSI, LSI मा logic function manufacturing बेला नै fixed हुन्छ।

तर technology विकाससँगै engineers ले आफ्नै आवश्यकता अनुसार hardware configure गर्न सक्ने devices develop भए, जसलाई **Programmable Logic Devices (PLDs)** भनिन्छ।

अर्कोतर्फ, large-scale commercial production को लागि **Gate Arrays** प्रयोग गरिन्छ जसले performance र cost दुवै improve गर्छ।

What is a Programmable Logic Device (PLD)?

Definition

A **Programmable Logic Device (PLD)** is an integrated circuit whose logic function can be programmed or reprogrammed by the user after manufacturing.

नेपालीमा,

PLD भनेको यस्तो IC हो जसको logic circuit factory मा permanently fix हुँदैन। User ले आफूलाई चाहिएको logic function अनुसार configure गर्न सक्छ।

Simple Example

Suppose एउटा factory लाई एउटा circuit चाहिएको छ जसले

Output = A AND B

अर्को factory लाई

Output = A OR B

चाहिएको छ।

Traditional IC मा दुईवटा अलग chips चाहिन्छ।

तर एउटा PLD प्रयोग गरेर programming मात्र परिवर्तन गरेर दुवै circuit बनाउन सकिन्छ।

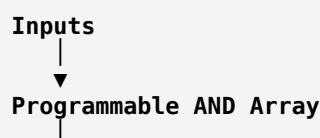
Why PLDs are Needed?

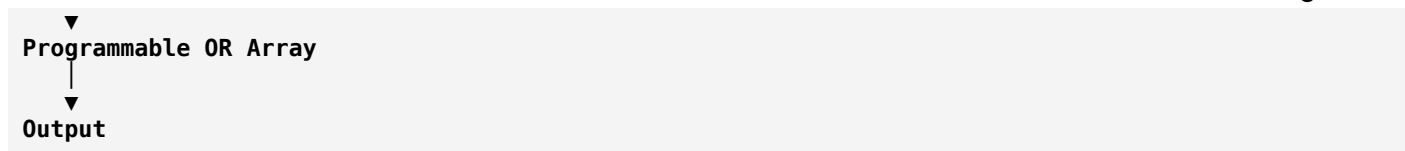
PLD development हुनुका मुख्य कारण:

- Flexibility
- Easy modification
- Short development time
- Lower design cost
- Reusable hardware
- Fast prototyping

Internal Structure of PLD

PLD मुख्य रूपमा तीन भाग हुन्छ।



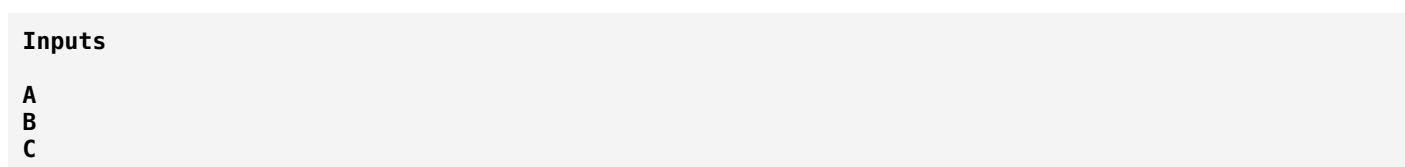


यसमा

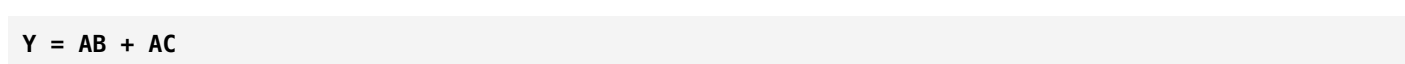
- Input signals प्रवेश गर्छन्।
- AND array मा product terms बनाइन्छ।
- OR array ले final logic बनाउँछ।
- Output प्राप्त हुन्छ।

Working Principle

Suppose

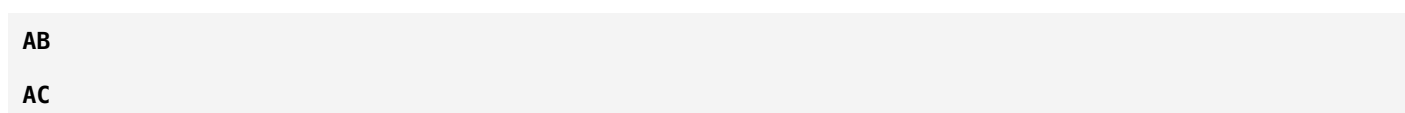


Need

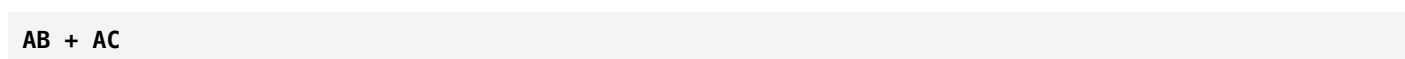


Programming गर्दा

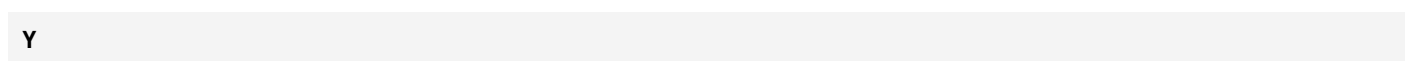
AND Array बनाउँछ



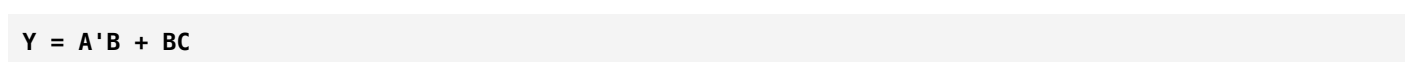
OR Array



Final Output



Programming change गरेपछि



समेत बनाउन सकिन्छ।

Hardware change गर्नु पर्दैन।

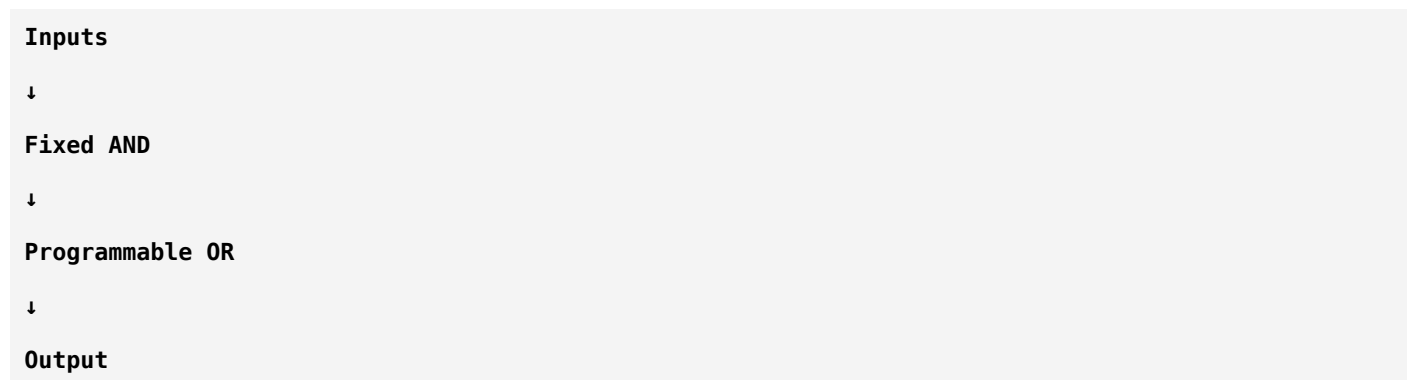
Types of PLDs

1. PROM (Programmable Read Only Memory)

PROM मा

- AND array fixed हुन्छ।
- OR array programmable हुन्छ।

Structure



Uses

- Lookup tables
- Fixed functions
- Code conversion

Advantages

- Cheap
- Simple

Disadvantages

- Limited flexibility

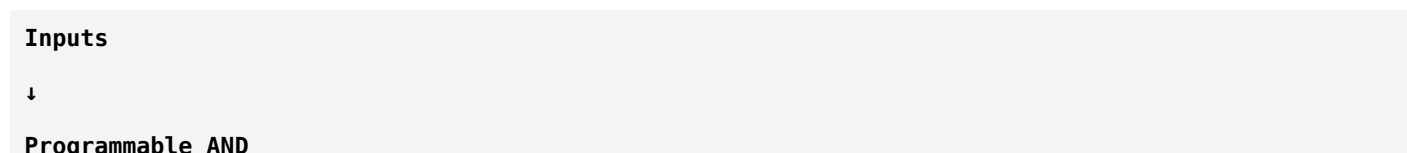
2. PAL (Programmable Array Logic)

PAL सबैभन्दा लोकप्रिय PLD हो।

यसमा

- AND array programmable
- OR array fixed

Structure



↓
Fixed OR
 ↓
Output

Advantages

- Fast
- Cheap
- Easy design

Applications

- Embedded systems
- Industrial control

3. GAL (Generic Array Logic)

GAL भनेको PAL को improved version हो।

Features

- Reprogrammable
- EEPROM based
- Multiple times programming

Advantages

- Reusable
- Flexible

4. CPLD (Complex Programmable Logic Device)

CPLD धेरै PAL हरू combine गरेर बनाइन्छ।

Structure

Multiple Logic Blocks
 ↓
Interconnection Matrix
 ↓
I/O Blocks

Features

- Large circuits
- Fast timing
- Non-volatile

Applications

- Automotive
- Robotics
- Industrial controllers

5. FPGA (Field Programmable Gate Array)

FPGA सबैभन्दा powerful programmable device हो।

यसमा हजारौं वा लाखौं logic blocks हुन्छन्।

Structure

Logic Blocks

Interconnect Network

I/O Blocks

Memory Blocks

Clock Management

FPGA लाई manufacturing पछि पनि unlimited design अनुसार configure गर्न सकिन्छ।

Applications

- AI accelerators
- Image processing
- Signal processing
- Video processing
- Networking
- Space systems

Advantages

- Extremely flexible
- Parallel processing
- Very high speed

Disadvantages

- Expensive
 - Higher power consumption
 - Complex design
-

Advantages of PLDs

1. Reprogrammable

Hardware फेरि बनाउनुपर्दैन।

2. Flexible

Same chip ले धेरै logic implement गर्न सक्छ।

3. Reduced Development Time

PCB redesign आवश्यक पर्दैन।

4. Lower Cost

Small production मा economical हुन्छ।

5. Easy Debugging

Programming change गरेर bug हटाउन सकिन्छ।

6. Easy Upgradation

Future features software बाट add गर्न सकिन्छ।

Disadvantages of PLDs

- High power consumption
 - Slower than ASIC
 - Expensive for mass production
 - Programming tools आवश्यक हुन्छन्
-

Gate Arrays

Definition

Gate Array भनेको एउटा **semi-custom integrated circuit** हो जसमा manufacturing समयमा धेरै logic gates पहिले नै तयार हुन्छन्।

पछि customer को requirement अनुसार interconnections मात्र बनाइन्छ।

नेपालीमा,

Gate Array मा gates factory मै तयार हुन्छन्। पछि wiring मात्र customize गरेर आवश्यक circuit बनाइन्छ।

Internal Structure

Prefabricated Gates

□□□□□□□□□□

□□□□□□□□□□

□□□□□□□□□□

↓

Customized Metal Connections

↓

Desired Circuit

Working Principle

Factory ले लाखौं gates भएको chip तयार गर्छ।

Customer ले भन्न सक्छ

Need:

Adder

Counter

Decoder

Processor

Manufacturer ले gates change गर्दैन।

केवल

Connections

मात्र बनाउँछ।

त्यसैले development धेरै छिटो हुन्छ।

Characteristics of Gate Arrays

- Semi-custom IC
- Faster than PLDs

- Less flexible
- Low manufacturing cost (large volume)
- High performance
- Low power consumption

Advantages of Gate Arrays

High Speed

Customized wiring भएकाले delay कम हुन्छ।

Lower Power

Power consumption कम हुन्छ।

Low Cost in Mass Production

Millions of chips उत्पादन गर्दा सस्तो पर्छ।

Reliable

Manufacturing quality धेरै राम्रो हुन्छ।

Disadvantages

- Reprogramming गर्न सकिँदैन।
- Design change गर्न नयाँ fabrication आवश्यक हुन्छ।
- Initial setup cost धेरै हुन्छ।
- Small projects का लागि उपयुक्त हुँदैन।

PLD vs Gate Array

Feature	PLD	Gate Array
Programming	User programmable	Factory customized
Flexibility	Very High	Medium
Reprogramming	Yes	No
Development Time	Very Short	Medium
Production Cost	High per chip	Low in large quantities
Speed	Medium	High
Power	Higher	Lower
Best For	Prototyping, research	Mass production

Real-Life Applications

Programmable Logic Devices (PLDs)

- Consumer electronics
- Robotics
- Industrial automation
- Communication systems
- Medical devices
- Digital Signal Processing (DSP)
- AI and machine learning hardware
- FPGA-based computer vision

Gate Arrays

- Smartphones
- Digital cameras
- Network switches
- Microcontrollers
- Automotive electronics
- Televisions
- Gaming consoles

PLD vs Gate Array (Easy Analogy)

Imagine **LEGO blocks**:

- **PLD** = Like LEGO. You can assemble, disassemble, and rebuild the structure whenever you want. It is highly flexible and ideal for learning, testing, and prototyping.
- **Gate Array** = Like a building made with cement. Once constructed, it is fixed and cannot be easily changed. However, it is stronger, faster, and more cost-effective for large-scale production.

Key Exam Points

- **PLD** is a programmable IC whose logic functions can be configured by the user after manufacturing.
- Common types of PLDs are **PROM, PAL, GAL, CPLD, and FPGA**.
- **FPGA** is the most advanced and flexible PLD, containing a large number of programmable logic blocks.

- A **Gate Array** is a semi-custom IC with pre-fabricated logic gates where only the interconnections are customized.
- **PLDs** are best suited for **prototyping, research, and low-volume production** because they are reprogrammable.
- **Gate Arrays** are preferred for **high-volume commercial products** due to their higher speed, lower power consumption, and lower cost per unit in mass production.